

BINARY BLITZ

Emerging & Future Technology • Grades 6–10 • AI Permitted

THE CONCEPT

A live coding challenge where the real problem is revealed only on the day. Participants use AI and their preferred development tools to turn an unfamiliar technology problem into a working solution under time pressure.

ZERO-DAY

The actual challenge is revealed on event day.

AI PERMITTED

Use AI as part of your development workflow.

DEVICE COMPULSORY

Bring a capable personal development device.

GITHUB MANDATORY

Every final submission must be uploaded to GitHub.

THE ZERO-DAY FLOW

01

REVEAL

Problem released live

02

BUILD

Research, code and develop

03

TEST

Run, debug and improve

04

COMMIT

Upload final work to GitHub

05

DEMO

Present the working solution

SAMPLE PROBLEMS

The following are examples of the style and difficulty of Binary Blitz challenges. They are NOT the competition problems. The actual challenge will be revealed on the day.

01 AUTONOMOUS DELIVERY ROBOTS

A delivery company is testing small autonomous delivery robots whose front sensors frequently lose accuracy when dust accumulates on the protective lens, causing the robot to slow down unnecessarily or miss obstacles. Create a program that monitors the sensor readings, detects when the lens is likely obstructed, and alerts the operator before navigation becomes unreliable.

02 SMART WI-FI CHANNEL SELECTION

Modern Wi-Fi routers automatically choose wireless channels, but nearby networks can interfere with each other and reduce connection speeds. Create a program that analyses the detected networks, identifies channel congestion, and recommends the best channel for the router to use.

03 AUTONOMOUS DELIVERY DRONE CHARGING

Delivery drones lose useful flight time when they return to a charging station with more battery remaining than necessary. Create a program that uses a drone's route, distance, battery consumption and delivery priority to determine when it should recharge and which deliveries it should complete before returning.

OPEN TOOL POLICY

Binary Blitz is designed around modern development practice: participants may use the tools that help them turn the revealed problem into a functioning solution.

PROGRAMMING

Any suitable programming language may be used, including Python, C, C++, Java, JavaScript, TypeScript, C#, Go, Rust and others.

EDITORS & IDES

VS Code, Visual Studio, PyCharm, IntelliJ IDEA, Cursor, Replit, Jupyter and other development environments.

OTHER DEVELOPMENT TOOLS

Frameworks, libraries, APIs, databases, data-analysis platforms, visualisation tools, public documentation and other software may be used where relevant to the challenge.

THE TOOL DOES NOT WIN. THE SOLUTION DOES.

AI TOOLS

ChatGPT, Claude, Gemini, GitHub Copilot, Cursor AI, Microsoft Copilot, Perplexity, Grok, DeepSeek and other AI assistants may be used.

AI IS A TOOL — THE RESULT IS YOUR RESPONSIBILITY.

MANDATORY GITHUB SUBMISSION

Every participant/team must maintain a GitHub repository and upload the final solution. The repository must contain the working source code, a README/instructions, the final build and an AI usage disclosure.

NO GITHUB REPOSITORY = NO VALID SUBMISSION

SUBMISSION MINDSET

Build a working solution, test it under pressure, upload it to GitHub, and be ready to demonstrate what you created.